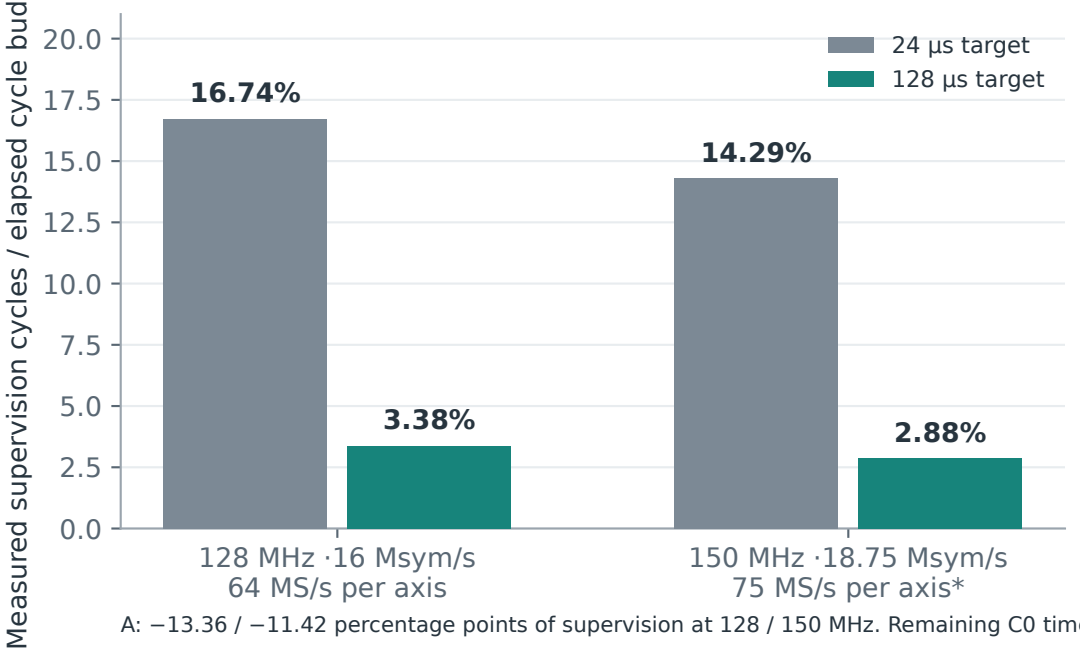
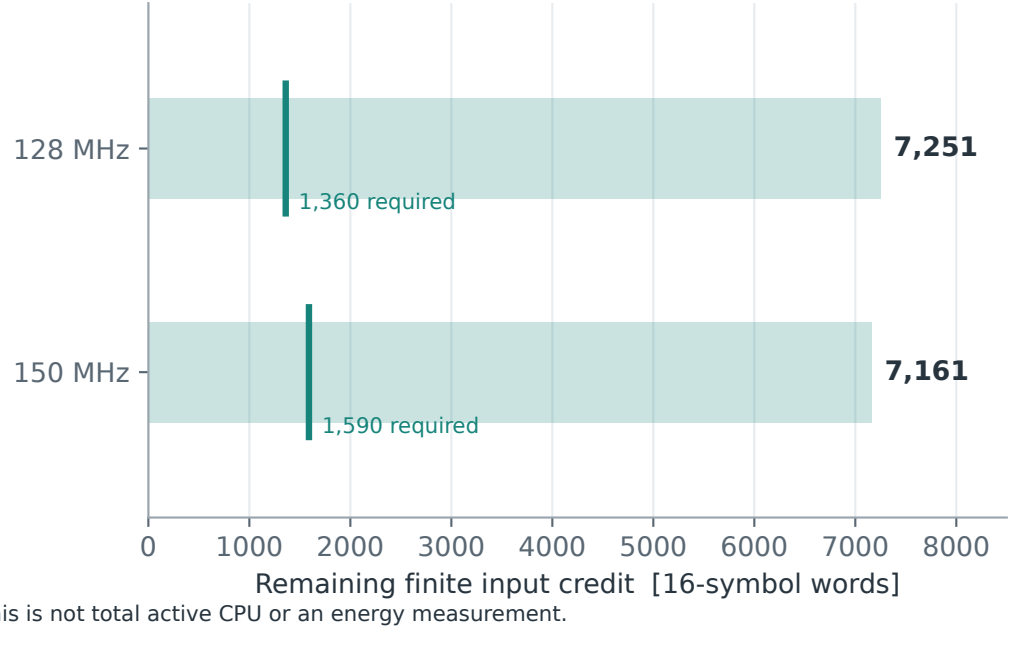
Hardware does the work; supervision can wait

Raw PRBS only · PIO pointers → FIFO-paced sample DMA · L8 / 4 SPS · Core1 idle · four clean 60 s runs

A Same raw source and output rate within each pair



B Instrumented 128 µs policy: minimum runway



A: −13.36 / −11.42 percentage points of supervision at 128 / 150 MHz. Remaining C0 time spins; this is not total active CPU or an energy measurement.

B: required runway 2B; B=8+ceil(fclk×257µs/49). Both-axis snapshot spans 138 / 136 µs ≤256 µs; cursor deadlines 3,132 / 2,673 µs.

Finite zero-credit sentinel stops unpublished reads. Old 24 µs runs did not record minimum runway or source-snapshot spans. All 4 captures: 8,192 observations, 0 mismatches.

*75 MS/s/axis exceeds the proposed 65 MS/s AFE limit. Digital GPIO evidence only; no FEC, SPI ingress, AFE or RF qualification.

Sources: quad-pointers/2026-10-15/{live-soak-1,poll128-soak-1}/runs.jsonl:1–2. Frozen validators, raw captures, bound formulas and hashes: waveform_selection.json.